

Mean-Variance Portfolio Optimisation: Analytical Implementation and Evaluation

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Contents

1	Introduction	3
1.1	Background	3
1.2	Problem Statement	3
1.3	Research Objective	3
1.4	Research Questions	4
1.5	Scope	4
2	Theoretical Background	5
2.1	Mean-Variance Optimisation	5
2.2	Portfolio Risk and Return	5
3	Methodology	6
3.1	Mathematical Formulation of the Problem	6
3.2	Analytical Solution	6
3.3	Numerical Quadratic Programming Solver	7
3.4	Experiment Design	7
3.4.1	Data Collection	7
3.4.2	Return and Risk Estimation	8
3.4.3	Portfolio Optimisation	8
3.4.4	Efficient Frontier Construction	8
3.4.5	Benchmark Portfolio	8
3.4.6	Performance Evaluation	9
3.4.7	Solver Validation	9
4	Results	10
4.1	Solver Validation	10
4.2	Efficient Frontier	11
4.3	Optimal Portfolio Allocation	12
4.4	Benchmark Comparison	12
5	Discussion	14
6	Conclusion	15

1 Introduction

1.1 Background

Portfolio construction is a fundamental problem in quantitative finance, where investors seek to allocate capital across multiple financial assets in order to achieve an appropriate balance between expected return and investment risk. Since individual assets exhibit different levels of expected return, volatility, and correlation, determining an optimal allocation is a challenging optimisation problem.

Mean-Variance Optimisation (MVO), introduced by Harry Markowitz, provides a mathematical framework for constructing portfolios that minimise investment risk for a desired level of expected return. By modelling risk using the covariance between asset returns, MVO demonstrates how diversification can reduce portfolio volatility without necessarily sacrificing expected performance.

Although modern optimisation libraries are capable of efficiently solving the MVO problem, understanding the underlying optimisation process remains valuable from both an educational and practical perspective. Furthermore, while the optimisation algorithm determines the optimal portfolio weights, the quality of the resulting portfolio ultimately depends on the estimated expected returns and covariance matrix derived from historical market data.

This project develops an analytical implementation of Mean-Variance Optimisation from first principles, validates the implementation against a numerical quadratic programming solver, and evaluates its out-of-sample portfolio performance using historical financial market data.

1.2 Problem Statement

Constructing an investment portfolio involves selecting asset allocations that appropriately balance expected return and investment risk. Simple allocation strategies, such as equally weighting all assets, fail to account for differences in asset volatility and correlation, potentially resulting in suboptimal portfolios.

Mean-Variance Optimisation addresses this problem by identifying the portfolio with the lowest possible risk for a specified expected return. While modern optimisation libraries provide efficient numerical solutions to this problem, it remains important to understand the underlying mathematical formulation and verify that independently implemented analytical solutions produce equivalent results.

This project investigates whether an analytical implementation of Mean-Variance Optimisation based on Lagrange multipliers produces equivalent portfolio allocations to numerical quadratic programming solvers, and evaluates the financial performance of the resulting portfolios using historical market data.

1.3 Research Objective

The objectives of this project are to:

- Develop a Mean-Variance Optimisation solver from first principles using the method of Lagrange multipliers.
- Validate the analytical implementation by comparing its solutions against industry-standard quadratic programming solvers.
- Construct optimal portfolios using historical asset return data.
- Evaluate the out-of-sample investment performance of optimised portfolios using historical market data.

- Compare optimised portfolios against common benchmark allocation strategies, such as equal-weight portfolios.

1.4 Research Questions

This project aims to answer the following research questions:

1. Does the analytical implementation produce equivalent portfolio allocations and objective values to a standard quadratic programming solver?
2. How does the optimal asset allocation change as the target portfolio return varies?
3. Does Mean-Variance Optimisation produce superior risk-adjusted performance compared with a naive equal-weight portfolio?

1.5 Scope

This project considers the classical unconstrained Mean-Variance Optimisation formulation with only target-return and fully invested constraints. Additional practical constraints, such as long-only positions, upper and lower allocation bounds, transaction costs, and turnover constraints, are outside the scope of this project. These constraints convert the optimisation problem into a more general constrained quadratic programming problem and eliminate the closed-form analytical solution derived using Lagrange multipliers. Consequently, they are left as potential extensions for future work.

2 Theoretical Background

2.1 Mean-Variance Optimisation

Mean-Variance Optimisation is a portfolio construction technique that seeks to minimise investment risk while achieving a specified level of expected return. The method assumes that investors prefer portfolios with higher expected returns and lower risk, where risk is measured by the variance of portfolio returns.

Unlike naive allocation strategies, Mean-Variance Optimisation explicitly considers both the individual volatility of each asset and the covariance between asset returns. This allows diversification effects to be incorporated into the portfolio construction process, potentially reducing overall portfolio risk.

By solving the optimisation problem for multiple target returns, the set of optimal portfolios forms the efficient frontier, representing portfolios that offer the highest expected return for a given level of risk.

2.2 Portfolio Risk and Return

The expected return vector is

$$\mu \in \mathbb{R}^n$$

where each element represents the expected return of an individual asset.

Similarly, the covariance matrix

$$\Sigma \in \mathbb{R}^{n \times n}$$

captures both volatility of each asset and the correlation between each assets.

Given portfolio weights

$$w \in \mathbb{R}^n$$

the expected portfolio return is

$$\mathbb{E}[R_p] = \mu^T w$$

while the portfolio variance is

$$\text{Var}(R_p) = w^T \Sigma w$$

These two quantities form the basis of the Mean-Variance Optimisation problem.

3 Methodology

3.1 Mathematical Formulation of the Problem

Given,

- Expected return vector,

$$\mu \in \mathbb{R}^n$$

- Covariance matrix,

$$\Sigma \in \mathbb{R}^{n \times n}$$

- Target return,

$$r \in \mathbb{R}$$

Find,

- Portfolio weighting vector,

$$w \in \mathbb{R}^n$$

That solves the following optimisation problem:

$$\begin{aligned} \min_w \quad & w^T \Sigma w \\ \text{s.t.} \quad & \mu^T w = r \\ & \mathbf{1}^T w = 1 \end{aligned}$$

The objective minimises portfolio variance while satisfying two equality constraints. The first ensures the portfolio achieves the desired expected return, while the second ensures that all available capital is allocated across the portfolio.

3.2 Analytical Solution

The optimisation problem is solved analytically using the method of Lagrange multipliers. Introducing multipliers λ and γ for the target-return and fully invested constraints respectively gives the Lagrangian,

$$L(w, \lambda, \gamma) = w^T \Sigma w - \lambda(w^T \mu - r) - \gamma(\mathbf{1}^T w - 1)$$

Take the gradient with respect to w and set it to 0,

$$\nabla_w L = 2\Sigma w - \lambda\mu - \gamma\mathbf{1} = 0$$

Rearranging gives

$$w = \frac{1}{2}\Sigma^{-1}(\lambda\mu + \gamma\mathbf{1})$$

Substituting this expression into the equality constraints produces a system of two equations in λ and γ , which could be solved numerically. Once these multipliers are obtained, the optimal weights are computed directly.

Rather than computing the inverse of the covariance matrix directly, the implementation uses Cholesky decomposition to improve numerical stability and computational efficiency.

3.3 Numerical Quadratic Programming Solver

To validate the analytical implementation, the same optimisation problem is solved using the `cvxpy` Python library, formulated as a standard quadratic program with linear equality constraints. The resulting portfolio weights, objective values, and computational performance are compared with those produced by the analytical implementation.

3.4 Experiment Design

3.4.1 Data Collection

Historical adjusted closing prices were obtained using the `yfinance` Python library. Daily price data covering the period from 1 July 2020 to 30 June 2026 were collected for fifteen financial assets spanning multiple asset classes.

To evaluate the ability of the optimisation model to generalise to unseen data, the dataset was divided chronologically into separate training and testing periods. Data from **1 July 2020 to 30 June 2025** were used to estimate expected returns and covariance matrices and to construct the optimal portfolios. The resulting portfolios were then evaluated on an out-of-sample testing period from **1 July 2025 to 30 June 2026** without re-estimating the model parameters.

The selected assets are summarised in the following table.

Ticker	Asset	Class
AAPL	Apple Inc	Equity
AMZN	Amazon.com Inc	Equity
BAC	Bank of America Corp	Equity
BND	Vanguard Total Bond Market ETF	Bond ETF
GLD	SPDR Gold Trust	Commodity ETF
JPM	JPMorgan Chase & Co	Equity
MSFT	Microsoft Corp	Equity
MSTR	Strategy Inc	Equity
NVDA	Nvidia Corp	Equity
PYPL	PayPal Holdings Inc	Equity
QQQ	Invesco QQQ ETF (Nasdaq-100)	Equity ETF
VNQ	Vanguard Real Estate ETF	Real Estate ETF
VOO	Vanguard S&P500 ETF	Equity ETF
VWO	Vanguard FTSE Emerging Markets ETF	International Equity ETF
VXUS	Vanguard Total International Stock ETF	International Equity ETF

Table 1: Selected assets

3.4.2 Return and Risk Estimation

Daily simple returns were computed from the adjusted daily closing price using

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

The expected return vector was estimated using the sample mean of daily returns, while the covariance matrix was estimated using the sample covariance of the return series. Both estimates were annualised assuming 252 trading days per year:

$$\mu_{\text{annual}} = 252 \times \mu_{\text{daily}}$$

$$\Sigma_{\text{annual}} = 252 \times \Sigma_{\text{daily}}$$

These annualised estimates form the inputs to both optimisation approaches. The annualised estimates computed from the training data remained fixed throughout the testing period to avoid look-ahead bias.

3.4.3 Portfolio Optimisation

The optimisation problem was solved using two independent approaches.

The first approach employed an analytical implementation derived from the Lagrangian formulation presented in [Section 3.2](#). The system of equations was solved using a Cholesky decomposition to avoid explicit matrix inversion.

The second approach solved the identical optimisation problem using a numerical quadratic programming solver. This implementation served as a reference solution against which the analytical implementation was validated.

For each target return, both approaches generated an optimal portfolio weight vector.

3.4.4 Efficient Frontier Construction

The efficient frontier was generated by solving the optimisation problem over a sequence of evenly spaced target returns spanning the feasible range of expected portfolio returns. For each target return, the analytical implementation computed the optimal portfolio weights and the quadratic programming solver computed the corresponding numerical solution. The resulting expected return and portfolio volatility were recorded. The collection of optimal portfolios formed the efficient frontier.

3.4.5 Benchmark Portfolio

To assess the effectiveness of Mean-Variance Optimisation, the optimised portfolios were compared against an equal-weight portfolio constructed using the same set of assets and evaluated over the identical out-of-sample testing period.

The equal-weight portfolio allocates

$$w_i = \frac{1}{n}$$

to every asset and therefore requires no optimisation. This benchmark provides a simple baseline against which improvements in portfolio risk and risk-adjusted performance can be evaluated.

3.4.6 Performance Evaluation

Portfolio values during the testing period were computed using a buy-and-hold strategy in which the initial portfolio weights obtained from the optimisation were fixed throughout the evaluation period.

The resulting portfolios were evaluated using the following metrics.

Expected Return

$$\mathbb{E}[R_p] = \mu^T w$$

Portfolio Volatility

$$\sigma_p = \sqrt{w^T \Sigma w}$$

Sharpe Ratio

Assuming a risk-free rate r_f ,

$$\text{Sharpe} = \frac{R_p - R_f}{\sigma_p}$$

The Sharpe ratio measures the expected excess return earned per unit of volatility. The risk-free rate R_f was assumed to be zero for the purposes of this evaluation.

3.4.7 Solver Validation

The analytical implementation was validated by comparing its output against the numerical quadratic programming solution.

The following quantities were compared:

- optimal portfolio weights
- portfolio variance
- expected portfolio return

Agreement between both approaches provides evidence that the analytical implementation correctly solves the Mean-Variance Optimisation problem.

4 Results

4.1 Solver Validation

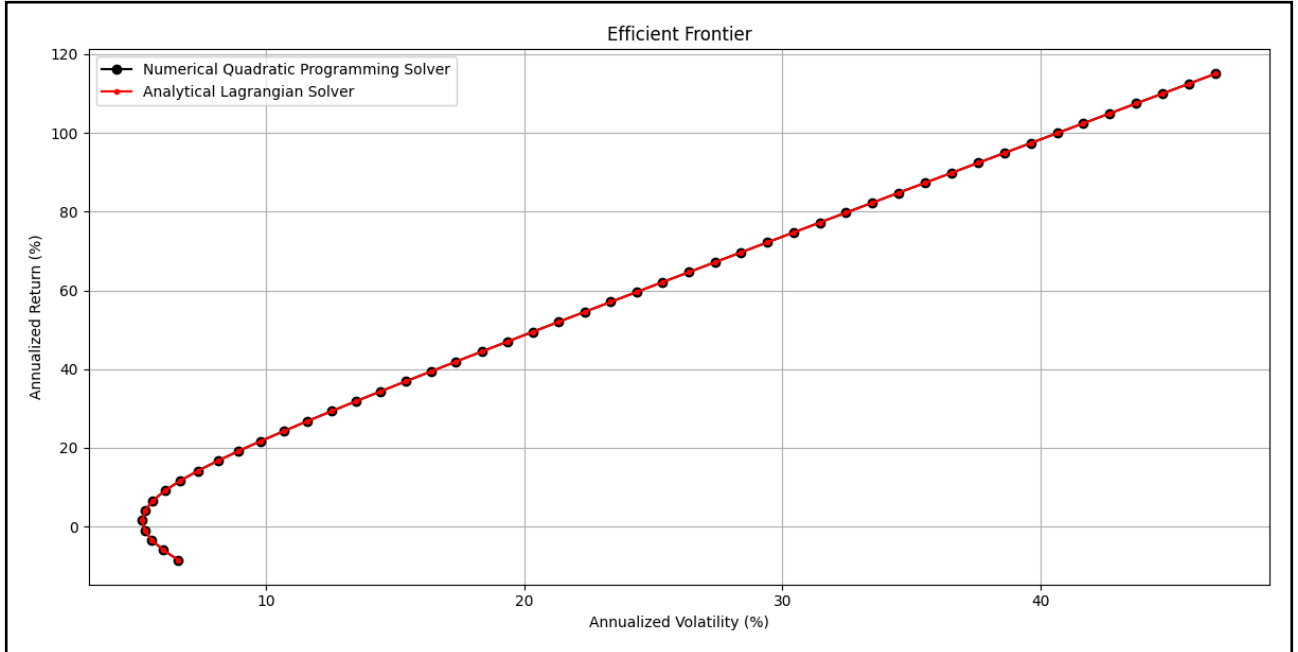


Figure 1: Efficient Frontiers of the two methods

The analytical implementation and the numerical quadratic programming solver were evaluated over fifty target portfolio returns spanning the feasible range of expected returns. For each target return, the portfolio weights, expected portfolio return, and portfolio variance obtained from both approaches were compared. Figure 1 shows that the efficient frontiers generated by the two methods are visually indistinguishable, indicating that both implementations produce the same sequence of optimal portfolios.

Statistic	Value
Number of target returns tested	50
Maximum weight difference	5.38×10^{-13}
Mean weight difference	4.87×10^{-14}
Maximum return difference	6.67×10^{-16}
Maximum variance difference	1.67×10^{-16}

Table 2: Overall validation results

Return	Max Weight Error	Return Error	Variance Error
10%	$< 1.1 \times 10^{-13}$	$< 1.3 \times 10^{-16}$	$< 4.4 \times 10^{-18}$
20%	$< 1.5 \times 10^{-13}$	$< 1.7 \times 10^{-16}$	$< 1.6 \times 10^{-18}$
30%	$< 1.9 \times 10^{-13}$	$< 5.6 \times 10^{-17}$	$< 7.0 \times 10^{-18}$
40%	$< 2.4 \times 10^{-13}$	$< 1.2 \times 10^{-16}$	$< 1.8 \times 10^{-17}$
50%	$< 2.7 \times 10^{-13}$	$< 1.2 \times 10^{-16}$	$< 1.4 \times 10^{-17}$

Table 3: Representative comparisons

Table 2 summarises the overall validation results, while Table 3 reports representative comparisons at selected target returns. Across all fifty optimisation problems, the maximum absolute difference in any portfolio weight was 5.38×10^{-13} , with differences in expected return and portfolio variance also remaining close to machine precision. These results demonstrate numerical agreement between the analytical implementation and the quadratic programming solver and validate the correctness of the analytical formulation.

4.2 Efficient Frontier

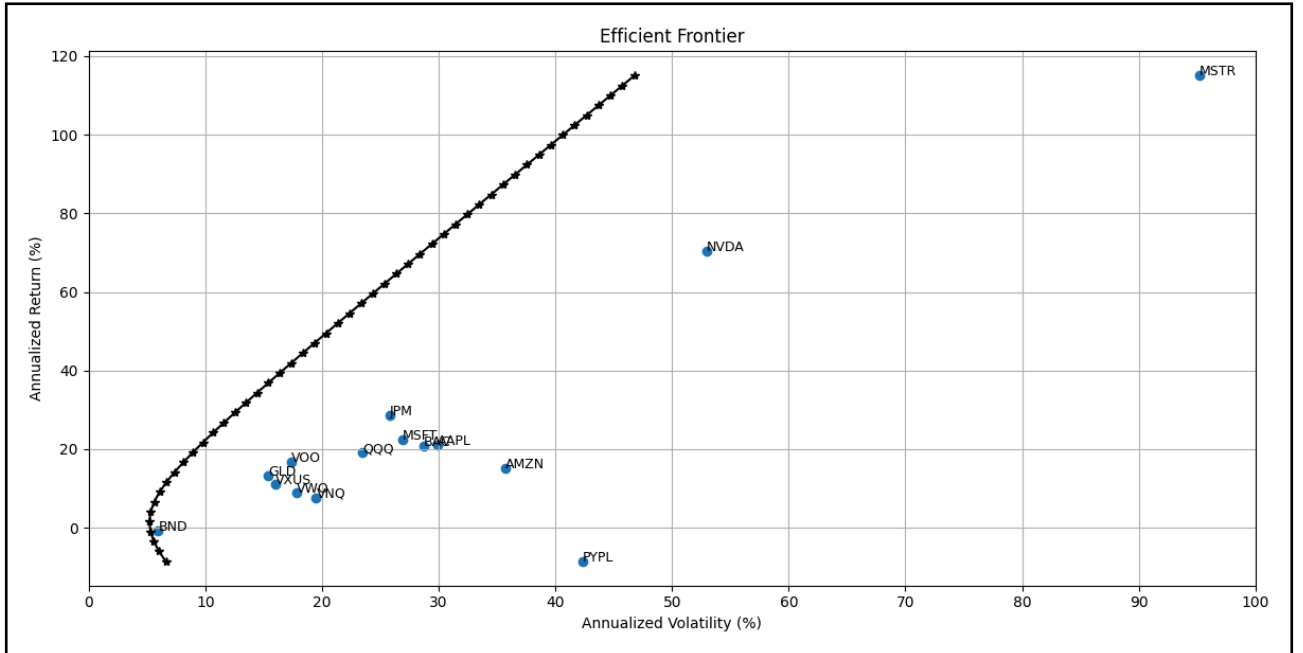


Figure 2: Efficient Frontier and individual assets

Figure 2 presents the efficient frontier together with the individual assets included in the dataset. Each point on the frontier represents the minimum-variance portfolio corresponding to a specified target return, while the scatter points represent the annualised risk and return of the individual assets. The frontier forms the upper boundary of feasible portfolios, illustrating the trade-off between expected return and portfolio volatility obtained through optimal diversification.

Several individual assets lie well below the efficient frontier, indicating that they are inefficient when considered in isolation. In contrast, diversified portfolios constructed through Mean-Variance Optimisation achieve higher expected returns for comparable levels of volatility or lower volatility for comparable levels of expected return, demonstrating the benefits of combining assets with different covariance structures.

4.3 Optimal Portfolio Allocation

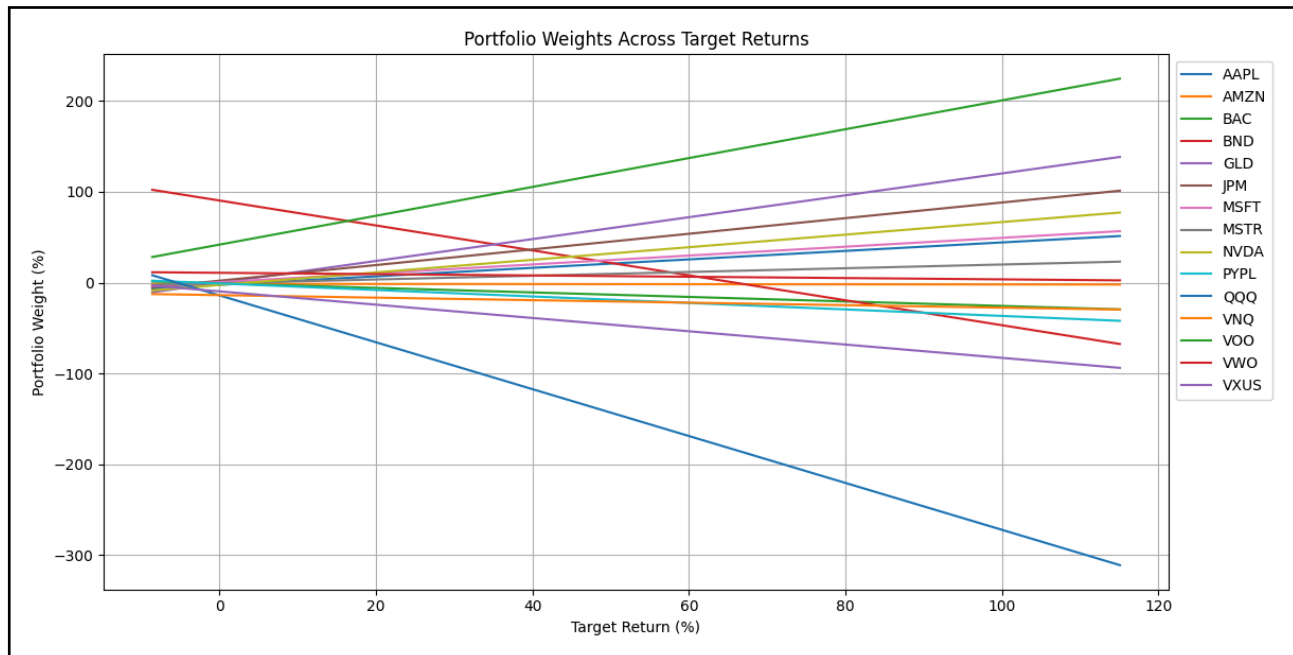


Figure 3: Optimal portfolio allocation across target returns

Figure 3 illustrates how the optimal portfolio weights evolve as the target return increases. As expected, the composition of the optimal portfolio changes continuously across the efficient frontier, with some assets receiving larger allocations while others decrease or become negative. Because the optimisation problem is unconstrained, negative portfolio weights correspond to short positions that are used to achieve the desired risk-return trade-off.

4.4 Benchmark Comparison

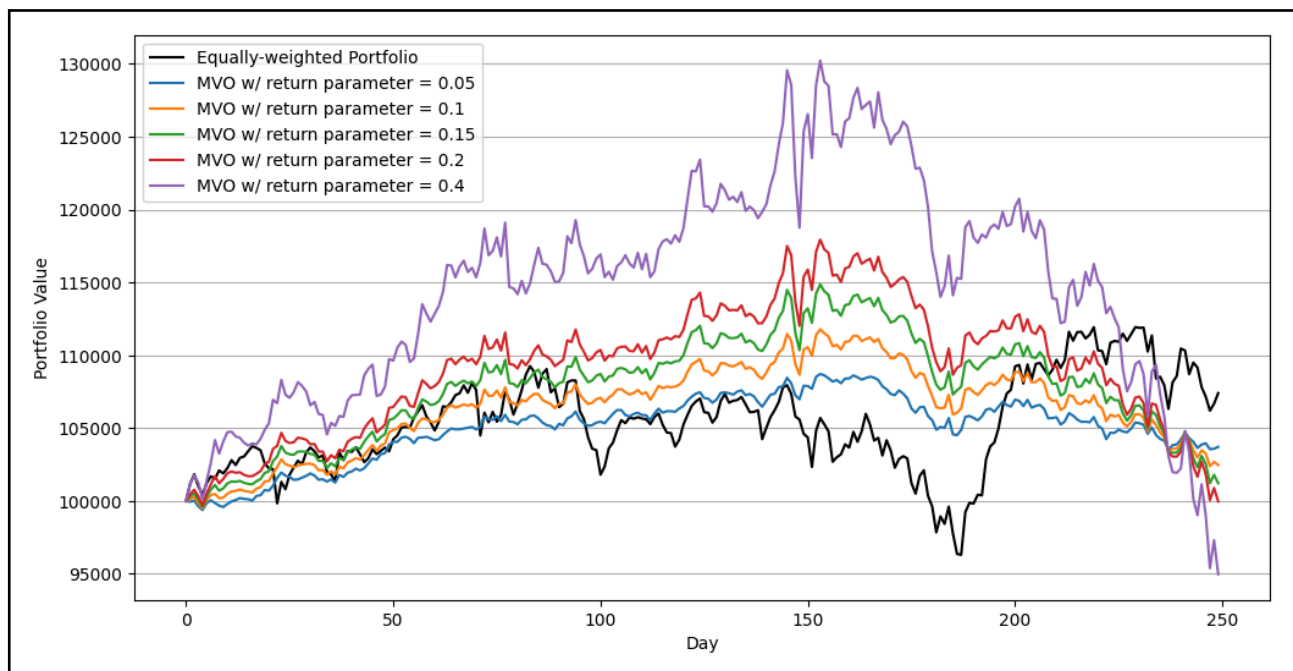


Figure 4: Portfolio value on different asset weighting

Strategy	Annual Return	Annual Volatility	Sharpe Ratio	Max Drawdown
Equally-weighted	7.4%	13.9%	0.53	−11.9%
MVO, return = 5%	3.7%	4.7%	0.79	−4.8%
MVO, return = 10%	2.5%	6.6%	0.38	−8.4%
MVO, return = 15%	1.2%	8.8%	0.14	−11.9%
MVO, return = 20%	−0.1%	11.1%	−0.00	−15.2%
MVO, return = 40%	−5%	20.4%	−0.25	−27.1%

Table 4: Performance evaluation metrics

Figure 4 compares the out-of-sample performance of several Mean-Variance Optimised (MVO) portfolios with different target returns against an equally weighted benchmark portfolio over the testing period. Each portfolio was constructed using the training data from 1 July 2020 to 30 June 2025 and subsequently evaluated over the independent testing period from 1 July 2025 to 30 June 2026 without re-optimisation.

The results indicate that portfolio performance depends strongly on the chosen target return. The conservative MVO portfolio with a target return of 5% exhibited the lowest volatility throughout the testing period and experienced substantially smaller drawdowns than the equally weighted benchmark. This behaviour is reflected in Table 4, where the 5% target portfolio achieved the highest Sharpe ratio (0.79) despite generating a lower annual return than the benchmark. Its maximum drawdown of only 4.8% further suggests superior downside risk management.

As the target return increased, portfolio performance deteriorated noticeably. Portfolios targeting returns of 20% and 40% exhibited larger fluctuations in portfolio value, deeper drawdowns, and ultimately negative annual returns over the testing period. Although these portfolios were expected to achieve higher returns according to the optimisation model, the realised performance was significantly worse. This discrepancy highlights the sensitivity of Mean-Variance Optimisation to estimation error in historical expected returns. Portfolios targeting aggressive returns tend to allocate heavily toward assets with the highest estimated historical returns, producing concentrated portfolios that generalise poorly when market conditions change.

Interestingly, the equally weighted portfolio achieved the highest realised annual return (7.4%) among the strategies evaluated while maintaining a Sharpe ratio of 0.53. Although it does not minimise portfolio variance, diversification across all assets provides robustness against estimation error. These findings suggest that, while Mean-Variance Optimisation can substantially improve risk-adjusted performance when conservative target returns are selected, aggressive optimisation may lead to inferior out-of-sample performance due to over-fitting of historical data.

5 Discussion

This project successfully developed an analytical implementation of the classical Mean-Variance Optimisation (MVO) problem and evaluated its performance using historical financial market data. The results demonstrate that the analytical solution derived using the method of Lagrange multipliers is mathematically equivalent to a standard numerical quadratic programming solver. Across fifty optimisation problems, the two approaches produced portfolio weights, expected returns, and portfolio variances that agreed to within machine precision, providing strong evidence that the analytical derivation and implementation correctly solve the unconstrained MVO problem. This successfully addresses the first research question regarding the correctness of the analytical implementation.

The efficient frontier and portfolio allocation results further illustrate the theoretical behaviour of Mean-Variance Optimisation. As the target return increases, the optimiser gradually shifts capital toward assets with higher estimated expected returns while increasingly utilising short positions to satisfy the return constraint. Because the optimisation considered the unconstrained formulation, this behaviour is consistent with the theoretical model and demonstrates how the portfolio composition evolves along the efficient frontier. This demonstrates how the portfolio composition evolves along the efficient frontier, addressing the second research question on how allocation changes with different target return.

The out-of-sample results show that the performance of Mean-Variance Optimisation depends strongly on the selected target return. The portfolio targeting a 5% annual return achieved the highest Sharpe ratio and the smallest maximum drawdown, indicating that the optimisation effectively reduced portfolio risk. In contrast, portfolios targeting higher expected returns performed progressively worse during the testing period, while the equally weighted portfolio achieved the highest realised annual return. These findings suggest that although the optimisation algorithm correctly identifies portfolios that are optimal with respect to the estimated model parameters, superior mathematical optimality does not necessarily translate into superior future investment performance. This therefore addresses the third research question, in which the answer depends on the target return chosen, rather than being a uniform yes or no.

This outcome highlights an important limitation of classical Mean-Variance Optimisation. The optimisation is highly dependent on the estimated expected returns and covariance matrix derived from historical data. When these estimates differ from future market behaviour, the resulting portfolios may generalise poorly to unseen data, particularly when pursuing aggressive return targets. Consequently, the benchmark results should not be interpreted as a failure of the optimisation algorithm itself, but rather as a reflection of the uncertainty inherent in estimating future asset returns.

Finally, this study considered the classical unconstrained formulation of Mean-Variance Optimisation in order to derive and validate the analytical solution. Practical portfolio management often incorporates additional constraints such as long-only positions, transaction costs, leverage limits, and periodic rebalancing, all of which were beyond the scope of this project. Incorporating these features, together with more robust estimation techniques, represents a natural direction for future work.

6 Conclusion

This project developed and validated an analytical implementation of unconstrained Mean-Variance Optimisation using Lagrange multipliers. The implementation produced solutions equivalent to those of a numerical quadratic programming solver and was successfully applied to historical financial data.

The evaluation demonstrated that although Mean-Variance Optimisation provides mathematically optimal portfolios for the estimated model parameters, realised investment performance depends heavily on the quality of those estimates. Conservative portfolios achieved superior risk-adjusted performance, while aggressive target-return portfolios generalised poorly during out-of-sample testing.

Overall, this project demonstrates both the theoretical strengths and practical limitations of classical Mean-Variance Optimisation. Future work may incorporate more realistic portfolio constraints and robust parameter estimation techniques to improve performance in real-world investment settings.